In [1]:
```
#Exp no.:7
```

In [2]:
```
#Aim: To perform data visualization using matplotlib
```

In [3]:
```
#Name:Anurag A Jagnade
#Roll:27
#sec:c
#subject:DSS
#Date:03-09-2026
```

In [5]:
```
import numpy as np
from matplotlib import pyplot as plt
```

In [6]:
```
x=np.arange(1,11)
```

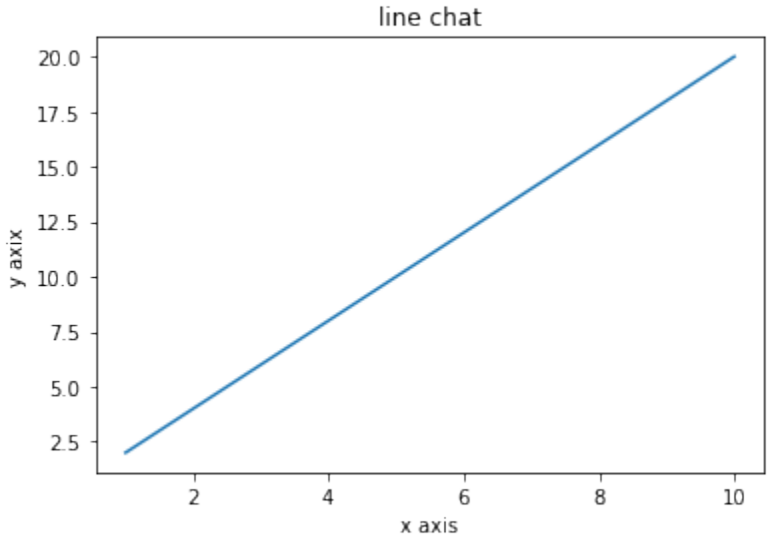
In [7]:
```
x
```

Out[7]:
```
array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10])
```

In [8]:
```
y=2*x
```

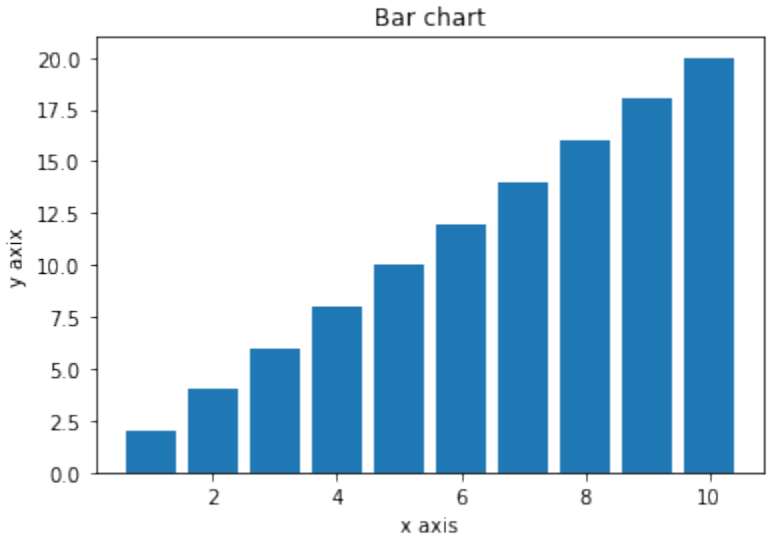
line chart

In [9]:
```
plt.plot(x,y)
plt.title("line chat")
plt.xlabel("x axis")
plt.ylabel("y axix")
plt.show()
```

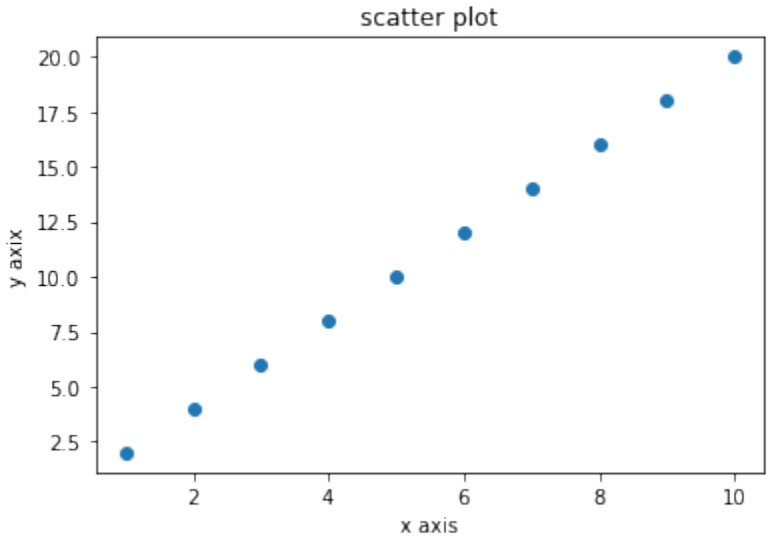


bar chat

In [10…
```
plt.bar(x,y)
plt.title("Bar chart")
plt.xlabel("x axis")
plt.ylabel("y axix")
plt.show()
```



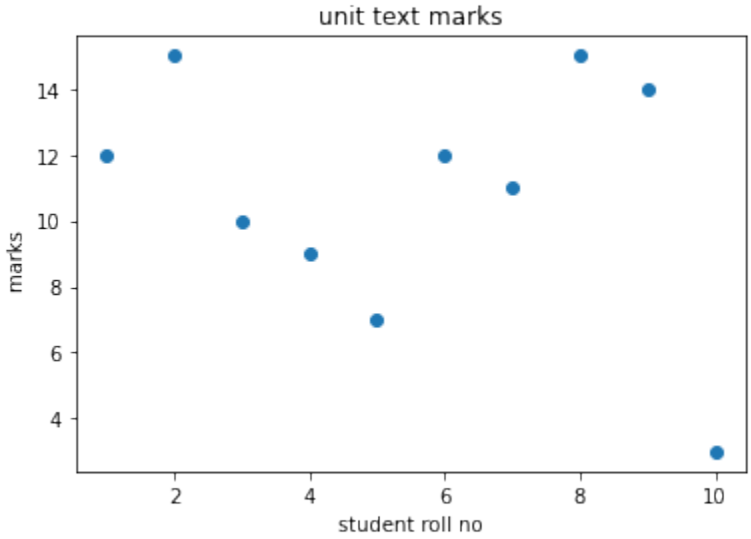
In [14…
```
plt.scatter(x,y)
plt.title("scatter plot")
plt.xlabel("x axis")
plt.ylabel("y axix")
plt.show()
```



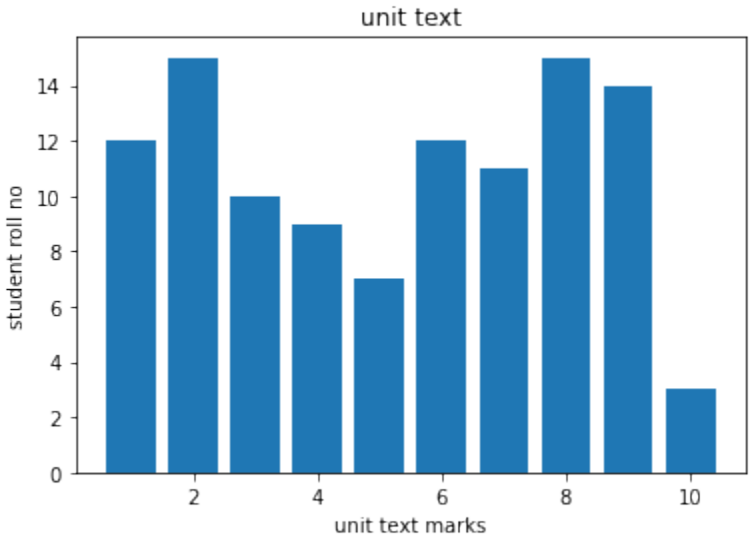
In [15…
```
a=(1,2,3,4,5,6,7,8,9,10)
b=(12,15,10,9,7,12,11,15,14,3)

plt.scatter(a,b)
plt.title("unit text marks")
plt.xlabel("student roll no")
```

```
plt.ylabel("marks  ")
plt.show()
```



In [25…
```
plt.bar(a,b)
plt.title("unit text")
plt.xlabel("unit text marks")
plt.ylabel("student roll no")
colors = ['red' if marks < 10 else 'blue' for marks in b]
plt.show()
```



In [26…
```
import matplotlib.pyplot as plt

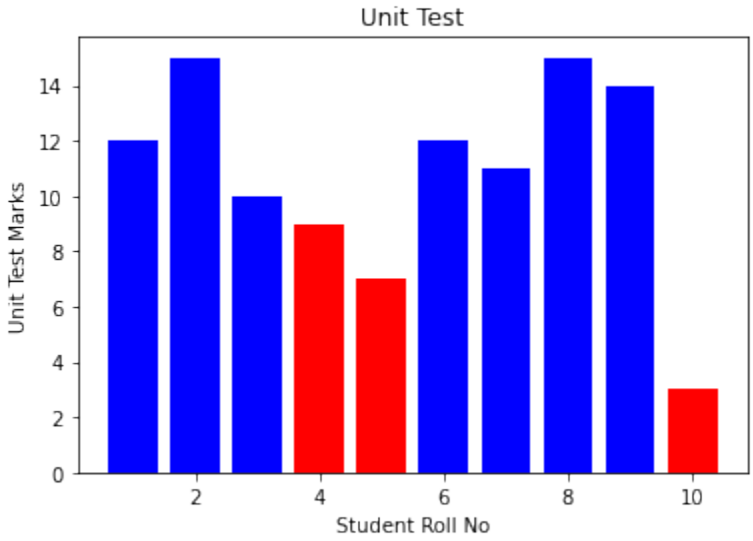
a = (1,2,3,4,5,6,7,8,9,10)
b = (12,15,10,9,7,12,11,15,14,3)

colors = ['red' if marks < 10 else 'blue' for marks in b]

plt.bar(a, b, color=colors)
```

```
plt.title("Unit Test")
plt.xlabel("Student Roll No")
plt.ylabel("Unit Test Marks")

plt.show()
```



In [ ]:

In [ ]: